

# Tristan Cravello

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• [www.linkedin.com/in/tristan-cravello-3b6500146/](https://www.linkedin.com/in/tristan-cravello-3b6500146/) • <https://github.com/triscravello> • <https://triscravello.github.io/tristan-portfolio-site/>

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## SUMMARY

Software Engineer with experience building full-stack applications, backend services, cloud-native infrastructure, AI-assisted engineering workflows, and distributed system prototypes using **TypeScript, Next.js, Python, React, Java, Spring Boot, PostgreSQL, Redis, Docker, Kubernetes, and AWS**. Experienced in **REST API design, CI/CD, containerized deployments, system reliability, AI-assisted debugging, and scalable architecture**. Former educator with strong communication, technical leadership, and problem-solving skills.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, SQL, HTML, CSS

**Backend:** Node.js, Express, FastAPI, Spring Boot, Spring Data JPA, REST APIs

**Frontend:** React, Next.js, Tailwind CSS

**Cloud & DevOps:** AWS, Docker, Kubernetes, GitHub Actions, CI/CD, Amazon ECS/Fargate, Amazon EKS, Amazon ECR, AWS Lambda, Terraform, OIDC

**Data & Distributed Systems:** PostgreSQL, PostGIS, MongoDB, Redis, Kafka, Prisma, Transactional Outbox, Idempotency, Event-Driven Architecture

**AI Engineering:** Retrieval-Augmented Generation (RAG), Claude Code, AI Agent Workflows, Model Context Protocol (MCP), Vector Search, Amazon Bedrock, Prompt Engineering

**Testing & Observability:** JUnit, Pytest, Testcontainers, Prometheus, Grafana, Kibana, CloudWatch, OpenAPI

## FEATURED PROJECTS

### [Threat Event Platform](#) | Cloud-Native Event Processing Platform

*Java • Spring Boot • Kafka • Elasticsearch • AWS ECS Fargate • Docker • GitHub Actions*

- Built a cloud-native threat event platform using **Java, Spring Boot, PostgreSQL, Kafka, and Docker** to ingest, process, and analyze security events through REST APIs.
- Integrated **Elasticsearch indexing** and Kibana dashboards to enable full-text threat search, filtering, and investigation workflows.
- Deployed to **AWS ECS Fargate** through **GitHub Actions OIDC**, Amazon ECR, Application Load Balancer, CloudWatch, and automated deployment rollback.

### [RunLayer](#) | Full-Stack Recommendation Platform

*Next.js • TypeScript • PostgreSQL • Prisma • Tailwind CSS*

- Built a modular recommendation engine combining weather, user preferences, and scoring algorithms to generate personalized running outfit recommendations.
- Implementing authentication, user profiles, recommendation history, saved outfits, and administrative gear management tools.
- Designed normalized **PostgreSQL** schemas supporting dozens of gear items, recommendation history, user profiles, and administrative workflows.
- Implemented comprehensive testing for recommendation logic, category diversification, personalization, and filtering behavior.

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## [Policy-Aware AI Repair Controller](#) | AI Engineering

*Python • Claude Code • Pytest • AI Agents*

- Built an AI-assisted repair workflow that diagnoses failures and proposes bounded code changes.
- Separated AI-generated repair decisions from deterministic policy enforcement.
- Restricted repair actions to explicitly permitted scopes and used automated tests as acceptance gates.
- Explored bounded-autonomy patterns for safer AI-assisted software engineering.

## [12 Million-Player Leaderboard](#) | Distributed Systems

*Python • Redis • AWS • Kinesis • Lambda • DynamoDB • ElastiCache*

- Designed a distributed leaderboard architecture for 12 million concurrent players.
- Partitioned rankings across **Redis** sorted sets to eliminate a single-key hotspot.
- Preserved exact global top-K ranking across distributed partitions.
- Designed a production AWS architecture using **Kinesis, Lambda, DynamoDB, and ElastiCache**.

## PROFESSIONAL EXPERIENCE

### Springboard, Remote

June 2025 - Present

*Software Engineering Fellow*

- Completed an intensive software engineering program focused on full-stack JavaScript development, software architecture, testing, databases, and API development.
- Built and deployed 15+ full-stack, cloud-native applications using **React, Next.js, Node.js, MongoDB, and PostgreSQL**.
- Collaborated on project planning, code reviews, debugging, and iterative feature development across full-stack applications.
- Applied modern development practices including Git workflows and continuous improvement.

### Discovery Academy of Science & Tinker K-8 School

July 2022 - May 2025

*Middle School Social Studies Teacher*

- Designed structured instructional systems and assessment workflows for 6th-8th grade World History, Civics, and United States History, serving **10-25** students across multiple learning tracks.
- Applied data-driven evaluation methods to monitor student performance trends and optimize instructional outcomes.

## EDUCATION

### University of South Florida, Remote

March 2026

*Software Engineering Bootcamp Certification*

### Florida State University, Tallahassee, Florida

April 2022

*Master's & Bachelor of Science in Social Science Education*